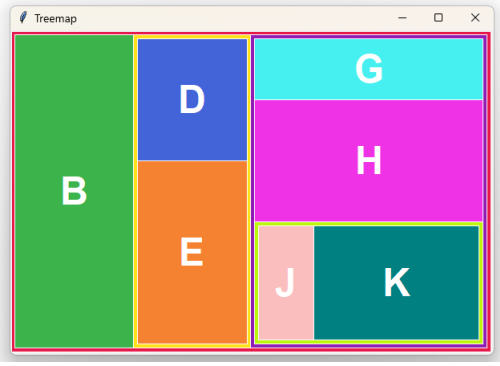
University of Trier
Software Engineering Group
Prof. Dr. Stephan Diehl

Information Visualization
Summer Semester 2026
Exercise Sheet 4

Group work: Please keep in mind that the exercises are meant to prepare you for the exam.
ChatBots: Use your own brain as much and as often as possible.

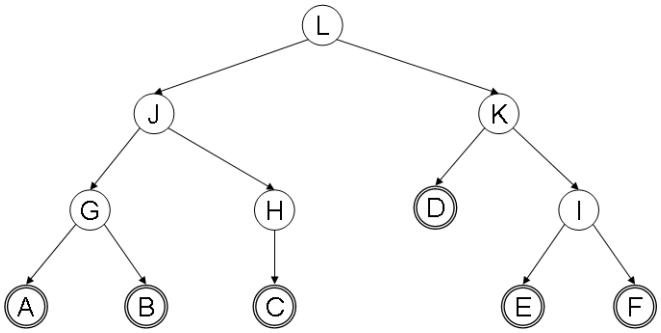
Exercise 1 (Implementation of a Treemap)

Implement a treemap in Java or Python. As a starting point for your implementation you should use the code provided in `exercise04_code.zip` in StudIP. The files contain an example tree with size information for each node. Your result should look similar to the treemap shown below:



Exercise 2 (Triangle Treemap)

A treemap can also be defined based on triangles instead of rectangles. The size of the triangles and sub-triangles must correspond to the size of the contained elements (leaf nodes), similar to the treemap described in Exercise 1. Assume that all leaf nodes have the same size. Draw a Triangle Treemap representing the below tree and explain how you separated the triangles. Tip: Reflect on how the area of a triangle can be separated into equally-sized sub-triangles.



If you enjoy programming, then implement your approach in Java or Python based on your solution of Exercise 1.